

Conversion of waste into wealth in chemical recycling of polymers: Hydrolytic depolymerization of polyethylene terephthalate into terephthalic acid and ethylene glycol using phase transfer catalysis

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ABSTRACT

Hydrolytic depolymerization of polyethylene terephthalate (PET) waste was studied using high pressure autoclave reactor at 240 °C (molten state) and autogenous pressure using excess of water in the presence of a phase transfer catalyst polyethylene glycol (PEG 400) and the product profile was traced at various time intervals. In comparison with zinc acetate (used before), PEG 400 was the best catalyst. Concentration profiles were developed for PET, oligomer and terephthalic acid (TPA) using HPLC. Effect of initial molar ratio of 22–110 mol of water/mol PET on depolymerization was studied by using both HPLC and end group analysis. Initial molar ratio of 55–110 mol of water/mol PET was found to give 99 percent conversion in 30 min at 240 °C. A complete conversion of ester linkages to acid group and the desired product TPA was studied. On the basis of ester linkage, 100% conversion was observed in 10–12 min reaction time. At 30 min additional time, it was found to give 95% yield of TPA. Molar concentration of PEG 400 of 2.0×10^{-5} mol/cm³ was sufficient to give the maximum conversion in breaking of ester linkages. The yield and purity of TPA was found to be 90 and 99.1%, respectively. A new mechanism of solid (polymer)-liquid (melt)-liquid (water) phase transfer catalysis (PTC) for hydrolysis was proposed and validated. A pseudo-first order rate equation was fitted for depolymerization with a rate constant of 1.4 min^{-1} at 240 °C and apparent activation energy of 34.4 kJ/mol. The rate of hydrolysis is very fast and complete depolymerization takes place within 30 min. This is an excellent example of circular economy and cleaner production from waste plastic.

1. Introduction

It is impossible to conceive of modern life without the use of plastics in one form or the other. The plastic market size in the world was valued at USD 609.1 billion in 2022 which is expected to grow at a compound annual growth rate (CAGR) of 3.7% up to 2030 (Grand view research, 2023). Plastics such as Polyethylene Terephthalate (PET) and Polycarbonates (PC) are employed in the packaging of apparel, beverages, appliances, consumer goods, toys, etc. (Cornago et al., 2021). PET is extensively used in food packaging, fibers, audiotapes, videotape, containers, etc. In particular, the packaging of appliances is likely to offer financial opportunities for market growth. The colossal demand results into massive plastic consumption, resulting in exorbitant amount of waste which is not properly disposed leading to environmental

degradation. Plastic recycling, both physical and chemical, has received tremendous attention for two reasons: pollution abatement and valorization of plastic waste which has become part of the circular economy (Ledford, 2015).

PET does not directly pose a hazard to environment and human beings (Heidbreder et al., 2019). It is totally recyclable and is the most widely recycled plastic globally ranging from 60 to 90% with India topping the list. PET mainly contains high aromatic compounds that are stable and inert, which hinder microbial degradation (Cordier and Uehara, 2019). However, there are various researchers who have reported microbial degradation of PET (Müller et al., 2001; 2005; Wei et al., 2016). Furthermore, enzymatic routes were studied to increase productivity, reaction kinetics and stability of enzyme at high temperature by Biundo et al. (2018) and Castro et al. (2019). Engineered depolymerase showed enzymatic depolymerization of PET to 90%

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Nomenclature

$[P]_{x,s}$	Polymer PET as solid mol/cm ³
$[P]_{x,org}(l)$	Polymer PET as organic liquid mol/cm ³
$[P]_{x,i}(l)$	Polymer PET as organic liquid at interphase mol/cm ³
$[PEG]_i$	PEG Concentration at aqueous and organic interface mol/cm ³
$[P - PEG]_{x,i}$	PET and PEG complex at interphase mol/cm ³
$[HOH - PEG]_{x,i}$	PEG water complex at interphase of aqueous and organic phase mol/cm ³
$[P]_{x-n,aq}(l)$	Water soluble oligomer fraction in aqueous phase mol/cm ³
$[P]_{n,i}$	Water insoluble oligomer fraction in aqueous phase mol/cm ³
$[OHCOR_1COOH]_{org}(s)$	Solid organic phase mol/cm ³
K_1	Distribution coefficient of PET between interphase and organic phase
K_i	Distribution coefficient of PEG in aqueous phase
K_3	Distribution coefficient of PEG in organic phase
C_{COOH}	Carboxyl group concentration at any time, mol/g polymer
C_{ELO}	Concentration of ester linkages in the polymer, mol/g polymer
X	Conversion of ester linkages to carboxylic group
k_{app}	Pseudo rate constant for hydrolysis, min ⁻¹

conversion in 10 h (Tournier et al., 2020). *Humicola insolens cutinase* (HiC) was used as an enzymatic catalyst for PET depolymerization into terephthalic acid (TPA), ethylene glycol (EG) and bis(hydroxyethyl) terephthalate (BHET) in 7 days at 70 °C (Brackmann et al., 2023). The foregoing reports not only show microbial catalytic routes for PET depolymerization but also suggest their potential to create a circular economy for polyester recycling. Therefore, an economic and eco-friendly process is needed to create a circular economy for the treatment of plastic waste. Plastic recycling presents a massive economic incentive to get feedstock and environmental protection due to the excessive consumption of non-renewable feedstock. In Europe, 4–6% of oil and gas is consumed for the production of polymer materials (King and Locock, 2022). Plastics should be part of the circular economy since 98% pure feedstock is used to generate 78 million tons of plastics; globally 72% of plastic is not recovered at all whereas 40% is landfilled and 30% is leaked to the environment. As mentioned before PET is the highest among all recycled plastics, in which 14% is used for energy generation (Raheem et al., 2019), 14% for cascade recycling and 2% for circular recycling (Cordier and Uehara, 2019). Quantities for recycled plastics are very low compared to the global recycling rate of paper (58%) and iron and steel (70–90%) (Graedel et al., 2011).

A new realization is that plastic waste should be considered as a resource rather than burden and innovative technologies and policies for recycle must be enacted by local governments. Polymeric waste can be recycled by four main approaches: *Primary Recycling*, *Mechanical Recycling* (Secondary), *Chemical Recycling* (Tertiary), and *Energy Recovery* (Quaternary) (Scheirs, 1998). The recycling techniques of primary and secondary, classified as mechanical recycling, vary in the amount of recycling material (Rios et al., 2007). Chemical recycling, upcycling and downcycling can be a good strategy to overcome the plastic waste menace. It is also termed as feedstock recycling which can further be categorised in three main approaches: depolymerization into its monomers, solvolysis, partial oxidation and cracking (thermal, catalytic (Lee et al., 2021), hydrogenation (Li et al., 2017), oxidation (Lovás et al., 2017), etc.).

Thermochemical methods and solvolysis of plastics are both

chemical recycling processes. Thermochemical process is carbon renewable technology and hydrolysis is polyester renewable technology. Both processes have their own limitations. Thermochemical process is better for dealing with carbon-carbon backbone polymer recycling. It would be better to design a system which is a combination of solvolysis and pyrolysis for mixed plastic (polyester + carbon-carbon backbone polymer) waste for carbon management. However, the thermochemical process requires high temperature (~500–700 °C) in comparison to solvolysis for breaking a polymer chain. In thermochemical process, the breaking of polymeric chain, whether random or selective, depends upon nature of catalyst. Upcycling of waste banner by thermocatalytic process using MnFeCoNiCu high-entropy alloy catalyst was done successfully. The yield of benzoic acid and phthalic acid was found to be 70–75% and 15–20%, respectively (Park et al., 2022a). Likewise, in catalytic pyrolysis of plastics, the product is distributed in gas, liquid and solid phases (Park et al., 2022b). The major limitation of thermochemical process is yield of desired product and demand for high energy. However, this process is helpful in recycling carbon-carbon backbone polymeric waste. On the other hand solvolysis process uses solvent as agent to break the plastic linkage. Solvolysis has a limitation to deal with carbon-carbon backbone polymeric waste. Treatment of nylon using thermochemical process yields the product which is a mixture of Caprolactam, CH₄, CO₂ and CO (Kim et al., 2022; Lee et al., 2022). However, it is convenient to convert ester into monomer by solvolysis with less product distribution. The recovered monomer can be further used for polyester synthesis.

Depolymerization is an axiomatic approach to green sustainability. Among these approaches, solvolysis is the most ideal, sustainable, green method for processing polymer waste. The mechanism involves chain scission of the polymer with the help of solvents. Solvolysis methods are named on the basis of the solvents used for depolymerization such as methanolysis (Liu et al., 2015), hydrolysis (Güçlü et al., 2003), glycolysis (Wang et al., 2015), and aminolysis (Tawfik and Eskander, 2010) and ammonolysis. These methods are extensively reviewed in the literature (Raheem et al., 2019) and effective depolymerization is carried out with supercritical or subcritical fluids; for instance, water or carbon dioxide (Liu et al., 2012), thermolysis (Sajdak and Muzyka, 2014), and ionic liquid (Wang et al., 2009). Most of the techniques consist of acidic (de Carvalho et al., 2006), alkaline (López-Fonseca et al., 2009) and expensive or hazardous chemicals (Sharma et al., 2013) using harsh reaction conditions. Depolymerization into monomer needs to possess desirable purity to be utilized in the re-polymerisation process. Neutral hydrolysis gives a mild separation process of product (monomer) and catalyst recovery. Depolymerization with volatile organic compounds can be responsible for the greenhouse gas (GHG) effect, depletion of ozone layer and global warming. Hence, depolymerization needs to be done by neutral hydrolysis (Breinbauer, 2007).

Polyethylene glycol (PEG) 400 is well known for widespread medical and industrial applications; it is soluble in polar and non-polar compounds. While PEG is generally used as a solvent, it can also be used as a catalyst in phase transfer catalysis (PTC) (Yadav and Motirale, 2009). Various phase transfer catalysts have been deployed in depolymerization, which may require less severe temperature conditions (Zhang et al., 2013). However, these phase transfer catalysts are uneconomical because of longer reaction times, non-reusability and environmental liability (De Lima-Borges et al., 2016). Most of PET hydrolytic depolymerization has been studied in the presence of metal acetate and ionic liquids (Al-Sabagh et al., 2014; Tincu et al., 2022). Depolymerization reaction above melt phase has been studied in reactive extrusion without catalyst. It shows 18–30% conversion in less than 10 min residence time. In reactive extrusion, the depolymerization conversion prominently depends upon temperature, residence time, pressure, water to PET feed ratio, etc. (Yalçinyuva et al., 2022). Also, alkaline hydrolysis with deep eutectic solvents was used for PET depolymerization in a microwave reactor, but the conversion of PET into TPA was found to be 80–82% (Attallah et al., 2021).

PET depolymerization using acidic, basic or neutral hydrolysis leads to several problems such as monomer instability in the reaction mixture, onset of secondary reactions and need for longer reaction time (de Carvalho et al., 2006; Liu et al., 2012). For example, in neutral hydrolysis, subcritical and supercritical depolymerization with zinc acetate is reported to cause dehydroxylation and dimerization of ethylene glycol (EG) (Sato et al., 2006). However, hydrolysis in the presence of PTC has no such undesirable reactions (Zhang et al., 2013). Depolymerization with PTC is mostly studied below melt phase, because of the lack of stability of ammonium salts based PTC at higher temperature. Although PTC-assisted depolymerizations have been conducted at lower temperature resulting higher yields, they suffer with excessive reaction time because these are solid-liquid (S-L) PTC reactions (López-Fonseca et al., 2009; Zhang et al., 2013).

In alkaline hydrolysis of PET, formation of salt with dimer or trimer of terephthalic acid (TPA) was found, which might have affected the yield of TPA (Karayannidis et al., 2002). As the breaking of a PET chain is due to a catalytic mechanism (Chen et al., 2013), a mixture with dimers or trimers of TPA may form sodium salt with NaOH (López-Fonseca et al., 2009). Consequently, it could be measured as TPA, as there is no separation between monomer and dimer. Most of the researchers have studied the extent of PET depolymerization based on a titration method (López-Fonseca et al., 2011). Indeed, in the case of complete depolymerization, this method may be applicable, but to study depolymerization behaviour, there is a need to have a reliable quantitative analysis such as HPLC and GC.

No research has appeared on using phase transfer catalysis for depolymerizing PET in neutral hydrolysis. In this work, a systematic analysis of the depolymerization of PET using polyethylene glycol (PEG) 400 as the phase transfer catalysis was done and kinetics of the process reported. It is an interesting case of multiphase transfer catalysis (Yadav, 2004, 2021). PEG 400 is stable under reaction condition unlike other ammonium or phosphonium based catalysts and is inexpensive, readily available and biodegradable. In addition, depolymerization based on ester linkages (carbonyl to carboxyl group) was studied and compared by HPLC analysis. In depolymerization, polymer chain scission is not necessarily ordered, but mostly random, i.e., formation of various oligomers. The formation of various oligomers limits the analysis process. A new method was thus developed for the analysis of oligomers and monomers. The overall process is an excellent example of circular economy through polymer (chemical) recycling.

2. Materials and methods

2.1. Material

PET flakes with sizes of 5–0.8 mm, were procured from Reliance Industries Ltd (RIL), Mumbai, India and PEG 400 from S. D.Fine Chemicals Pvt. Ltd., Mumbai. High purity HPLC grade water was used. All other chemicals used were of analytical grade obtained from reputed vendors.

2.2. Experimental

A stainless-steel autoclave (Amar Equipments, Mumbai, India), with internal diam. 75 mm, volume 300 mL, standard 4-blade pitch turbine impeller, equipped with proper agitation, heating arrangements and controllers, was used for all hydrolysis experiments. The reactor was preheated to 80–90 °C and the reactants added. The control experiment was done by using 111 mol H₂O/mol PET, 0.0021 mol of PEG 400/cm³. Typical experiments were done at 240 °C. The melting temperature of the resin was calculated to be 240 °C by differential scanning calorimetry (DSC). The polymer melt phase played an important role and the time required for the polymer to melt entirely was governed by the initial water to PET molar ratio (W/P) charged to the reactor. Complete melting occurred at 240 °C which was achieved in 25–35 min depending

on the W/P ratio (Fig. S1). Once the melting occurred, it was taken as the zero time. Since periodic sampling was not possible, independent experiments were done for different time intervals for the same W/P ratio to study the kinetics. After the desired time interval, heating was stopped, the reactor removed and quenched quickly in an ice bath. The reaction mass temperature dropped immediately to 100 °C within a minute of quenching. It would solidify the unreacted polymer, if any. Subsequently, the reactor was opened, and the product removed. The product was separated into two-phases (solid and liquid) using a sintered glass filter. No further washing was performed with cold water to remove any residual water-soluble components, such as ethylene glycol (EG) monomer. The aqueous phase composed of EG, water, PEG 400. EG presence was identified by Agilent 6890 GC (Agilent Technologies, USA) with a DB-5 capillary column (0.25 μm × 0.25 mm × 30 m with 5% diphenyl/95% dimethyl polysiloxane packing) and an FID (flame ionization detector). The GC method was set starting at 70 °C for 2 min with a ramp rate of 10 °C/min until 210 °C with a H₂ flow rate of 30 mL/min and N₂ as a carrier with a flow rate of 30 mL/min. The injection volume of the sample was 2 μL. The quantification of EG and presence of PEG 400 in liquid was done through HPLC-ELSD analysis. The recovered solid was dried at 80 °C under vacuum to a constant weight. The solid was ground using a mortar and pestle into a fine powder for analysis.

2.3. Qualitative analysis of decomposed PET

Solid samples were recovered from hydrolytic reaction at various time intervals and used for the thermal behaviour analysis by DSC and TGA (Perkin Elmer STA 6000) and XRD (Bruker AXS diffractometer, Cu-K radiation of wavelength 154 nm, at gazing angle (10°) 2θ scan). The weight loss and percent crystallinity from 30 to 350 °C with heating and cooling cycle with rate of 10 °C per min in nitrogen atmosphere provided the thermal history of material. The mass spectra of the products were obtained on an ISQ single quadrupole MS (Thermo Scientific) instrument with electron ionization (EI). Fourier transform infrared spectroscopy (FTIR) of samples was performed with Perkin Elmer spectra BX spectrophotometer in KBr pellets between 4,000 and 300 cm⁻¹. CP/MAS ¹³C NMR was conducted to investigate changes in the structure of decomposed PET using a JEOL JNM-ECA 500 spectrometer with an NM-93030 CPM probe. The solvent used for ¹³C NMR and ¹H NMR was DMSO-*d*₆ and the results were compared with the standard TPA pattern.

2.4. Quantitative analysis

The foremost difficulty in hydrolytic depolymerization of PET is to find a method for quantifying the extent of hydrolysis as a function of reaction time. Previous literature on evaluating the extent of hydrolysis covers intrinsic viscosity, infrared analysis, end group analysis and gel permeation chromatography (GPC). However, these techniques are satisfactory for PET hydrolysis below the melt phase (Golike and Lasoski, 1960) since complete depolymerization quantification includes fragmented oligomers to final monomer and unreacted polymer. PET recycling using DMSO and hydrotalcite was studied by (Sharma et al., 2013) to produce a wide range of oligomer formation wherein their structural characterization were achieved by mass spectrometry with complementary ¹³C NMR analysis. Solvent system used for calculation was required to be such that non-polar and polar oligomers become soluble. The identification of the main product of hydrolysis was done using HPLC with reversed phase C-18 column. The mobile phase was a mixture of acetonitrile: water (0.1% trifluoroacetic acid)-20:80 with flow rate of 1.0 mL/min. Detection was done using UV detector at 280 nm. The solid sample was dissolved in DMSO and diluted using methanol. TPA was confirmed by matching commercial TPA residence time (RT). A standard curve was used for HPLC analysis.

$$\% \text{ yield of TPA}(x) = \frac{n_{TPA}}{n_{TPA0}} \times 100 \quad (1)$$

Where, n_{TPA0} is theoretical concentration (in M (mol L^{-1})) and n_{TPA} is terephthalic acid concentration at any time (M). n_{TPA0} is calculated from degree of polymerisation of the polymer flakes used for experiments.

$$n_{TPA0} = C_{P0} \times n \quad (2)$$

Where, C_{P0} is PET initial concentration in M and n is degree of polymerisation.

In HPLC, two peaks were observed for reaction samples drawn. After more than 3 min whereas the sample collected after 30 min showed only one peak. The first peak was identified as TPA and second peak as oligomer. The complementary analysis using MS-EI showed the molar mass of reaction samples (at 3, 10, 15 min) 166 and 211–311 Da.

HPLC and end group analyses were done and compared because they correlated with scission of ester linkages, which was related to the kinetics of hydrolytic depolymerization as shown in Scheme 1.

The reaction extent was determined by the analysis of the solid phase for carboxyl group content. Titrimetric method was used for the two-solvent system depending upon solid product solubility, i.e. extent of hydrolysis, as described elsewhere (Khalaf and Hasan, 2012; Campanelli et al., 1993). Each chain scission gives one carboxyl end group, and hence measuring carboxyl group present at various reaction times gives the extent of depolymerization. The conversion of ester linkages calculated by using following relation:

$$\% \text{ conversion of ester linkages (Carbonyl to carboxyl)}(x) = \frac{C_{COOH}}{C_{ELO}} \quad (3)$$

Where, C_{ELO} is initial ester linkage concentration (mol/g polymer) and C_{COOH} is carboxyl group concentration at any time (mol/g polymer).

The statistical values are reported by taking a mean of three readings of conversion from three acid value for a specimen. Standard deviation in conversion was calculated using following equation:

$$\text{Standard deviation} = \sqrt{\frac{\sum (\bar{x}_i - x)^2}{n - 1}} \quad (4)$$

Where, $\bar{x}_i$ is average conversion and x is conversion at any time.

The liquid phase was analysed for EG and PEG 400 by HPLC and HPLC-ELSD respectively, using an Agilent BDS C8 column ($4.6 \text{ mm} \times 250 \text{ mm}$) with DI water (100%) mobile phase at flow rate of 0.6 mL/min with RI detector. The yield of ethylene glycol was calculated as follows:

$$\% \text{ yield of EG}(x) = \left(\frac{n_{EG}}{n_{EG0}} \right) \times 100 \quad (5)$$

Where, n_{EG0} is theoretical ethylene glycol concentration (M) and n_{EG} is ethylene glycol concentration at any time (M).

3. Results and discussion

Structural identification of products was done by using different

analytical methods.

3.1. Structural identification of products of hydrolysis of PET using PEG 400

The structural identification of main product (solid phase TPA) in the degradation of PET in water catalysed by polyethylene glycol 400 was performed by using different structural characterization tools such as DSC, TGA, FTIR, XRD, NMR, EI-MS.

3.1.1. DSC

Thermal analysis of the product received for various reaction time was performed with heating rate 10°C/min . The thermal analysis of samples studied by DSC is depicted in Fig. 1 which shows glass transition temperature (75°C), crystallisation temperature (165°C), and melting temperature (240°C) for samples at different times; the variation in these properties is clearly seen. Table S1 (supplementary information, SI) provides the thermal properties and change in crystallinity of PET samples at various reaction time. As the reaction time increases glass transition temperature disappears, which proves the complete scission of the polymeric chain at 30 min. The cleavage of polymeric chain after 15 min reaction time shows onset temperature of melting and peak temperature as 110.9 and 150°C , respectively, which are identical with that for BHET. After 15 min reaction time, there was an absence of endothermic peak showing an amorphous nature of the material. After 30 min a sublimation curve was seen which was similar to that of terephthalic acid. Melting endotherm gives an idea about the structure of polymer; as the reaction time increases the endothermic peak broadens and shifts toward lower melting temperature. Broadening of endothermic peak increases the enthalpy of heat of sublimation and it decreases crystallinity (Lucchesi and Lewis, 1968; Bi et al., 2016). It was attributed to terephthalic acid, as is seen for reaction at 30 min (Fig. 1).

3.1.2. TGA

The TGA profile for reactant and products at various reaction times

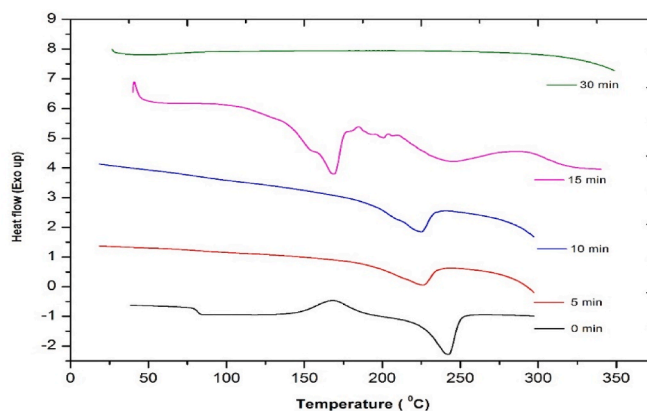
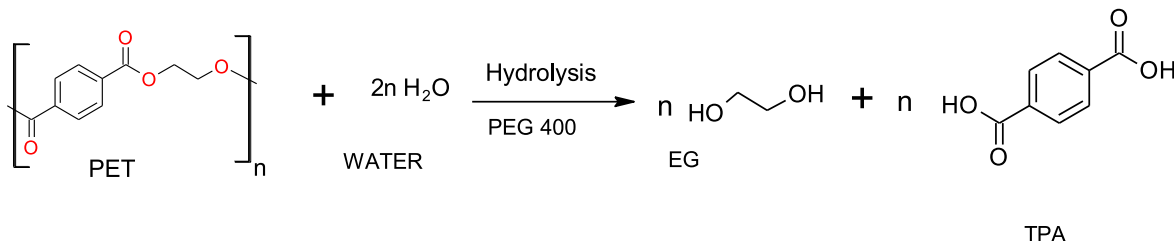


Fig. 1. DSC plot of PET hydrolysis catalysed by PEG-400 at different times.



Scheme 1. Reaction scheme for PET hydrolysis.

are shown in Figs. S2 and S3. The weight loss of ~92% in the range of 390–400 °C for zero min can be seen, which is caused by the thermal decomposition of PET. The first range of 200–300 °C shows nearly 40–50% weight loss, for 5–15 min reaction time. For the second range of 390–400 °C, a weight loss in the range of 60–50% was observed, which is attributed to thermal decomposition PET produced in TGA process (Zhou et al., 2012). For the third range of 230–340 °C, a weight loss of nearly 94% can be seen for 30 min reaction time. The third range of temperature is attributed to terephthalic acid, and sublimation occurs before the decomposition temperature (Elmas Kimyonok and Ulutürk, 2016). A similar trend is followed by PEG 400 hydrolytic product for 30 min reaction time.

3.1.3. XRD

XR diffractogram for different reaction samples is shown in Fig. S4 (supplementary information, SI); due to high degree of crystallinity of PET it exhibits a typical pattern, with broader peaks at $2\theta = 16.4, 17.7$, and 23.0 . The crystallinity of PET is different from crystallinity of terephthalic acid; progressive decrease in molecular weight of polymer can be seen in Fig. S2 due to breakage of polymer. The JCPDS (Pattern: 00-031-1916) data show excellent match with that of terephthalic acid.

3.1.4. FTIR

Fig. S5 shows FTIR of standard TPA and PET reaction conditioned with PEG 400 at different reaction times; PET shows a typical molecular vibration at 1734 (C=O stretch, ester), $1,238$ C–O, 1043 (–O–(CH₂)–O–), and 969 cm^{–1} C–O peaks of PET. FTIR shows carbonyl peak for all OH-stretch that keep increasing with increase in reaction time. As reaction time increased there was absence of ethylene glycol. These results are akin to the standard terephthalic acid (Mancini et al., 2010) which confirm that product as terephthalic acid.

3.1.5. ¹H NMR

The ¹H NMR for samples catalysed by PEG 400 for 30 min reaction time had the following shifts shown in Fig. 2 at δ 8.21 ppm. These spectra when compared with standard TPA are identical. After 10–15 min reaction the spectra indicates the presence of BHET, which is confirmed by the presence of signal for methylene protons of CH₂–OH, methylene group near –COO–, hydroxyl group at δ 3.9, 4.4, and 4.1 ppm, respectively. When glycolysis of PET with mixed oxides is carried out in the presence of hydrotalcite, it formed BHET monomer with 50 min at 196 °C (Chen et al., 2013). However, in our work using PEG 400 as a phase transfer catalyst, the hydrolysis of PET led to the formation of BHET monomer along with terephthalic acid within 10–15 min.

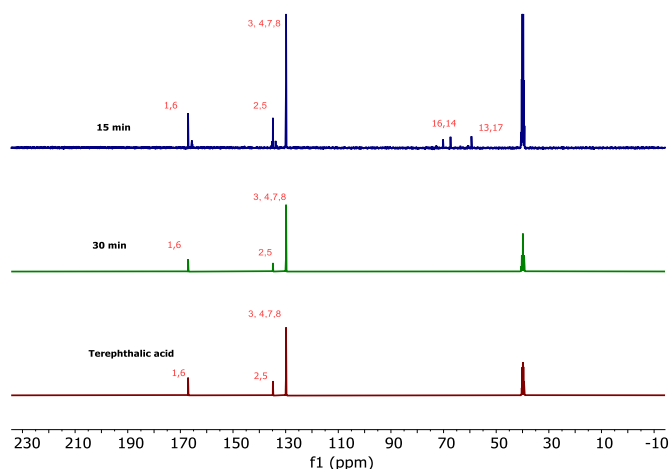


Fig. 2. ¹H NMR spectra of standard TPA and PET with PEG 400 at different reaction times.

3.1.6. ¹³C NMR

In NMR ¹³C spectra for standard reference spectra for TPA and that for the hydrolysis of PET with PEG 400 at different reaction times as shown in Fig. S6, wherein the signal at δ 129 ppm indicates aromatic protons of benzene ring, 167 COOH, and 134 aromatic C adjacent to ester. However, for 15 min reaction time it shows similar spectra; in addition spectra of methylene carbon of COO–CH₂ and CH₂OH indicated signals at 4.963 and 3.73 ppm, respectively. Also, reference spectra for the commercial terephthalic acid when compared with 30 min reaction sample is identical which confirms that the product formed is TPA. ¹³C NMR spectra of the hydrolysed product also suggests the purity of product formed in reaction is 99.1%.

3.1.7. MS

The structural characterization of long chain high molecular weight molecules can be achieved by mass spectrometry (MS) (Montaudou and Lattimer, 2001), which is complementary to NMR analysis as a reliable structural identification tool. In recycling of PET with DMSO solvent and hydrotalcite catalyst, Sharma et al. (2013) obtained an oligomer, whose molecular weight was analysed by MS. In the current work, with PEG 400, the MS shown in Fig. S7 shows molecular weight of 216 and 166 gm.mol^{–1}, which are identical to molecular weight of BHET and TPA for reaction times of 15 and 30 min.

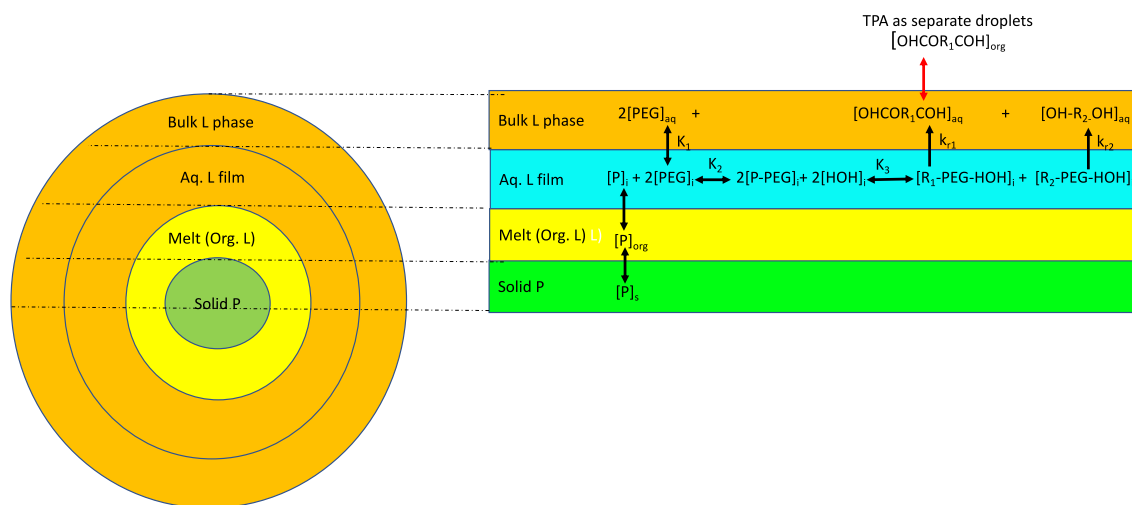
3.2. Reaction mechanism: S-L-L-L PTC

The reaction proceeds by solid (polymer)-liquid(melt)-liquid (water) phase transfer catalysis (PTC) as delineated in Scheme 2 at 240 °C. It ultimately becomes L(aq)-L (interfacial film)-L(org) PTC when all polymer is melted. When the temperature is lowered the liquid TPA becomes solid, EG remains in the aqueous phase along with PEG 400 making it possible to recycle the catalyst (see Scheme 3).

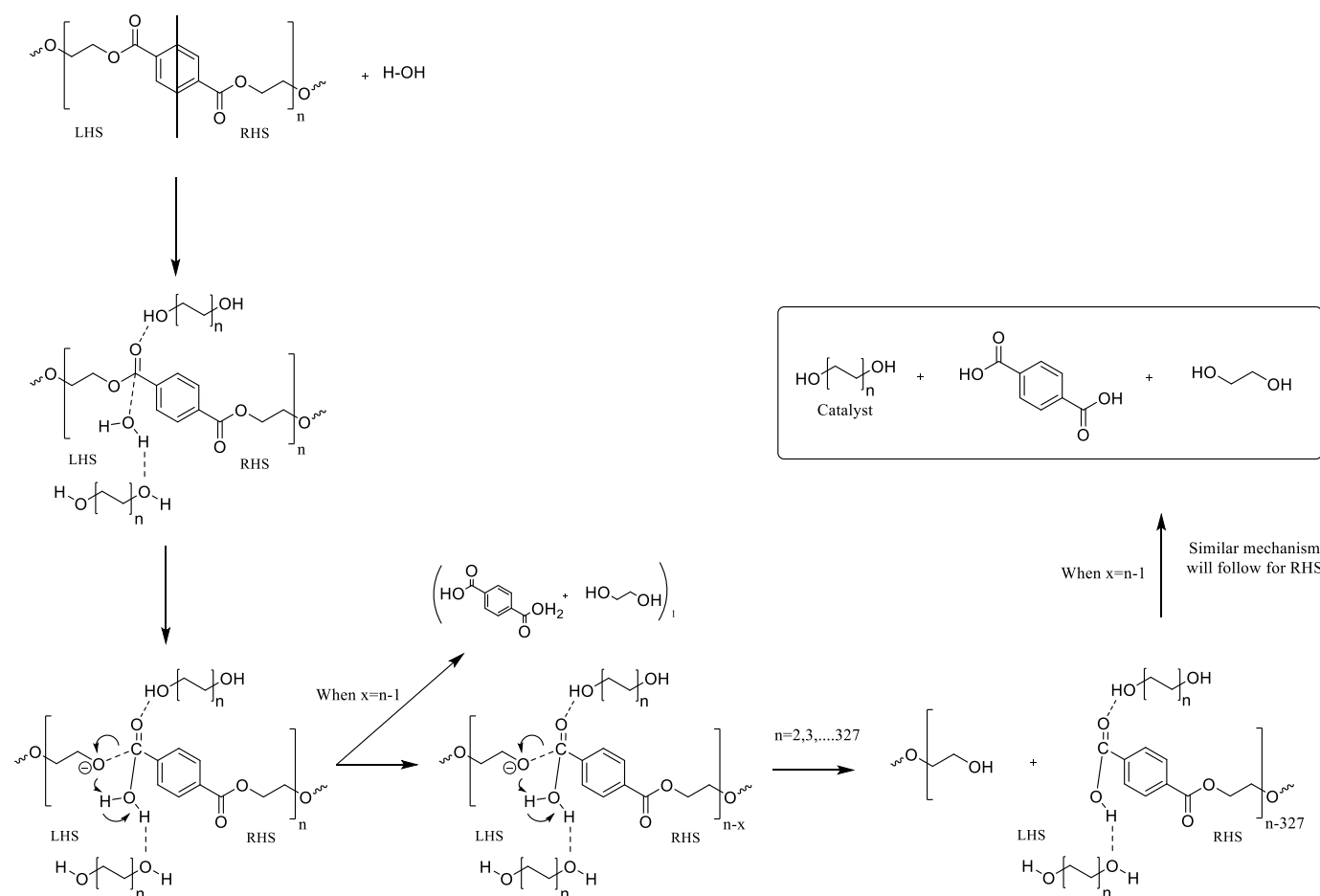
Various mechanisms of PTC for L(aq)-L(org), S(reactant)-L(org), L(aq)-L(middle)-L(org), S(cat/reactant)-L(aq)-L(org) and S-L(omega)-L(org) have been discussed and modelled by Yadav, 2004, 2021. The PET hydrolysis forms an interesting case of S-L-L-L PTC. PEG forms a hydrogen bonding with hydroxyl hydrogen of PEG 400 with carbonyl oxygen of ester linkage and reacts with PEG 400. Instantaneously PET long chain molecules scission takes place randomly. Oligomers with lower molecular weight is characterized by higher solubility in water with high dielectric constant. Such oligomers are then transferred to the aqueous phase, where the hydrolysis takes place, and immediate formation of terephthalic acid takes place.

Oligomers are characterized as insoluble in water and form a hydrogen bonding with PEG 400 and transferred to aqueous phase. Furthermore, this active intermediate is easily attacked by protonated water molecule because high dissociation energy of water at 240 °C. Consequently, all oligomer breaks down into monomer and there is a formation of TPA and EG. When all solid PET is in a molten state, it becomes L(melt)-L(interfacial film)-L(aq) PTC. Now the mechanism in the current case can be discussed and modelled.

As discussed before PET melts at 240 °C and hence the particles are softened from 75 °C towards the melt phase and there are 4 phases in the reaction mass: polymer PET [P]_s as a solid, softened and later converted into a melt as organic liquid [P]_{org}, a thin layer of aqueous film surrounding the organic phase which contains the phase transfer catalyst [PEG]_i and [P]_i which forms a complex [P-2PEG]_i and reacts with water in that phase. The catalyst is distributed between the aqueous phase and interfacial film. The polymer is hydrolysed with 2 mol of water to break it into two portions, giving rise to TPA (as liquid) and EG in the bulk aqueous phase. TPA is transferred out of the film as a separate organic phase and remains as a disperse droplets (Scheme 2). The catalyst in the film is in equilibrium with that in the bulk aqueous phase (equilibrium constant K₁). The solid P goes into a molten phase at 230 – 240 °C. Before the molten phase, P is sparingly soluble in the interfacial film and its rate is increased when the entire P is in the molten state. P form a complex



Scheme 2. General mechanism for PET depolymerization in the presence of PTC.



Scheme 3. Proposed catalytic mechanism for PET in presence of PEG 400.

with 2 mol of the catalyst PEG and the complex P-2PEG then reacts with 2 mol of water forming a complex P-2PEG-2HOH which is broken into two polymers complexes R_1 -PEG-HOH (acid part) and R_2 -PEG-HOH (glycol part) which dissociate into TPA and EG and transferred to the aqueous phase. TPA has practically no solubility in water and hence is transferred to the organic phase as a droplet. EG remains in the bulk

aqueous phase. The catalyst is regenerated.

The overall reaction for PET depolymerization in the presence of PEG is as follows:



PET solid particles are dispersed in the liquid phase containing water

in large excess and phase transfer catalyst (PEG). When the temperature is raised to melt PET, the solid particles are softened and converted in to globules containing exterior of molten PET and interior of solid PET. Ultimately, the PET droplets will be form within the aqueous phase. Then it becomes L-L PTC. The ester linkages of PET are broken according to the mechanism as given below.

$$[P]_x(s) \xrightleftharpoons{K_1} [P]_x(l) \quad (6)$$

$$K_1 = \frac{[P]_{x,org}}{[P]_{x,s}} \quad (7)$$

PET distribution in thin aqueous film (inter-phase)

$$[P]_{x,org}(l) \xrightleftharpoons{K_2} [P]_{x,i}(l) \quad (8)$$

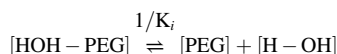
$$K_2 = \frac{[P]_{x,i}}{[P]_{x,org}} \quad (9)$$

3.2.1. PET forming complex with PEG

$$[P]_{x,i}(l) + [PEG] \xrightleftharpoons{K_3} [P-PEG]_{x,i} \quad (10)$$

$$K_3 = \frac{[P-PEG]_{x,i}}{[P]_{x,i}[PEG]} \quad (11)$$

When PEG is used as catalyst following complex may formed bonding with water molecule



$$K_i = \frac{[HOH-PEG]_{x,i}}{[PEG][H-OH]} \quad (12)$$

PEG distribution in various complexes is given by:

$$[PEG]_0 = [HOH-PEG]_{x,i} + [P-PEG]_{x,i} + [PEG]$$

$$[PEG]_0 = K_i[PEG][H-OH] + [P]_{x,i}[PEG] + [PEG]$$

The concentration of free PEG can be obtained from above equation

$$[PEG] = \frac{[PEG]_0}{[1 + K_i[H-OH] + [P]_{x,i}]} \quad (13)$$

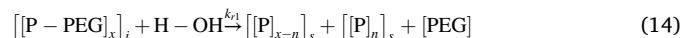
$$+ \frac{d([HOOCR_1COOH])_s}{dt} = k_{r3}K_i[P]_{n,i} \frac{[PEG]_0}{[1 + K_i[H-OH] + [P]_{x,i}]} [H-OH]_i[H_2O] \dots \quad (21)$$

Complex (active reactant) reacts with water in thin aqueous film (interphase) and breaks down PET in two fractions as follows:

1. Water soluble fraction (oligomer formation)
2. Water insoluble fraction (solid oligomers of higher molecular weight)

Breaking of PET is an instantaneous reaction. It was observed that there was no presence of unreacted PET in less than 3 min. This indicated that PET breaks down instantaneously in to mixture of oligomer as

mentioned above (water soluble and insoluble).



$$-\frac{d([P-PEG]_{x,i})}{dt} = k_{r1}[P-PEG]_{x,i}[H_2O] \quad (15)$$

1. The oligomer fraction which was readily water soluble reacts with water and is converted into monomer due to a very fast reaction. It was confirmed by the analysis of reaction mixture which showed the presence of monomers (TPA and EG).



$$-\frac{d([P]_{x-n})}{dt} = k_{r2}[P]_{x-n}[H_2O] \quad (17)$$

2. The solid oligomer fraction which is sparingly soluble in water slowly reacts with water and converts into monomer which is the rate determining step (RDS).



$$+ \frac{d([HOOCR_1COOH])_s}{dt} = k_{r3}[P]_{n,i}[H-OH-PEG]_i \quad (19)$$

The overall rate of reaction is given by Eq. (19).

The concentration of [H-OH-PEG] is unknown, and it is assumed that concentration of [P-PEG] to be the same as the concentration of oligomer found experimentally by HPLC analysis. The solid samples (intermediate) were collected and dissolved in DMSO at different reaction times for analysis.

The measured concentration of solid has been used for determination of kinetics (Fig. S8 HPLC analysis).

Substitution values of unknown concentration of [H-OH-PEG] from equation (12)

$$+ \frac{d([HOOCR_1COOH])_s}{dt} = k_{r3}[P]_{n,i}[H-OH-PEG]_i[H_2O] \quad (20)$$

$$+ \frac{d([HOOCR_1COOH])_s}{dt} = k_{r3}K_i[P]_{n,i}[PEG][H-OH]_i[H_2O]$$

The concentration of water is excess and substituting value for catalyst concentration

$$k_{app} = k_{r3}K_i \frac{[PEG]_0}{[1 + K_i[H-OH] + [P]_{x,i}]} [H-OH]_i[H_2O] \quad (22)$$

Substituting value for k_{app} equation (21) becomes

$$+ \frac{d([HOOCR_1COOH])}{dt} = k_{app}[P]_{n,i} \quad (23)$$

The concentration of PET can be written as $[P]_{n,t} = [P]_{n,0} - [OHCOR_1COH]$

Solving equation (23) for time- 0 to t

$$+\frac{d([[\text{HOOCR}_1\text{COOH}]])}{dt}=k_{app}[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]]_s \quad (24)$$

$$+\frac{d([[\text{HOOCR}_1\text{COOH}]]])}{[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]]}=k_{app}dt \quad (25)$$

Integrating equation (25) for limit time t = 0 to t = t

$$\int_{[[\text{HOOCR}_1\text{COH}]]_0}^{[[\text{HOOCR}_1\text{COH}]]_t} + \frac{d([[\text{HOOCR}_1\text{COOH}]]])}{[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]]}=k_{app}\int_0^t dt. \quad (26)$$

$$\ln\left(\frac{[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]_0]}{[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]_t]}\right)=k_{app}t \quad (27)$$

Where conversion can be defined as

$$X=\left(\frac{[[[\text{HOOCR}_1\text{COOH}]]_t]-[[[\text{HOOCR}_1\text{COOH}]]_0]}{[[P]_n]_0-[[[\text{HOOCR}_1\text{COOH}]]_0]}\right) \quad (28)$$

3.2.1.1. Assuming initial concentration of terephthalic acid = 0, then

$$X=\left(\frac{[[[\text{HOOCR}_1\text{COOH}]]_t]}{[[P]_n]_0}\right). \quad (29)$$

$$-\ln(1-X)=k_{app}t \quad (30)$$

$[[\text{HOOCR}_1\text{COH}]]_t$ = Concentration of TPA (recovered in solid phase was measured by dissolving into DMSO and analysed in HPLC (Fig. S8 and Fig. S9).

$[[P]_n]_0$ = Initial concentration of PET (mol/cm³)

$[[[\text{HOOCR}_1\text{COOH}]]_0$ = Initial concentration of TPA (mol/cm³)

3.2.2. Hydrolysis product analysis

Fig. 3 shows concentration profiles for PET, TPA and oligomer whereas Fig. S10 shows the yield of TPA. To investigate the effect of water concentrations on the apparent rate constant, experimental data were collected at 240 °C for PEG concentration of 2.11×10^{-5} mol/cm³ for reaction times up to 10 min. Fig. 4 shows data for rate constant calculated using TPA and end group analysis, the difference between two rate constants indicates that end group analysis measures acid group of TPA and dimer or trimer. As predicted the pseudo first order model fits the initial rate data at 240 °C and catalyst concentration 2.11×10^{-5}

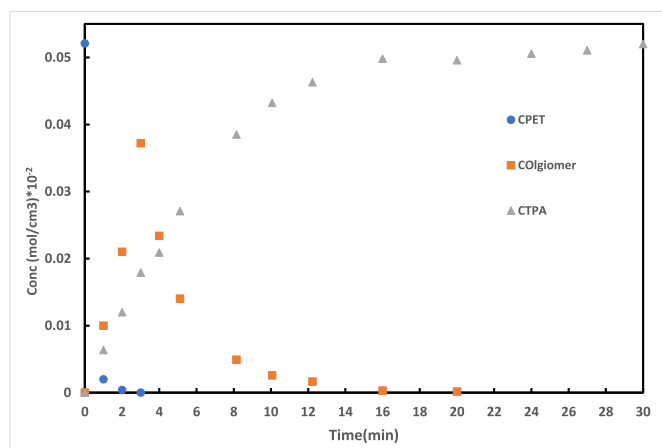


Fig. 3. Concentration profile for PET, TPA, intermediate oligomer with respect to time at 240 °C with PEG 400 catalytic effect. (Reaction condition- Temperature: 240 °C; Catalyst concentration: 2.11×10^{-5} mol/cm³; PET concentration: 5.27×10^{-4} mol/cm³; Water: 100 cm³; Speed of agitation: 650 rpm) (Analysis based on HPLC Method).

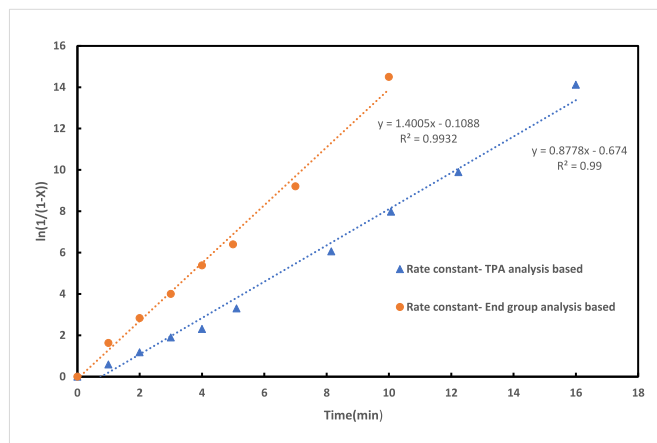


Fig. 4. Initial rate data based on TPA and end group analysis. (Reaction condition- Temperature: 240 °C; Catalyst concentration: 2.11×10^{-5} mol/cm³; PET concentration: 5.27×10^{-4} mol/cm³; Water: 100 cm³; Speed of agitation: 650 rpm).

mol/cm³, giving reaction rate constant 1.40 min⁻¹. The pseudo first-order kinetics gives a slope of 1.40 min⁻¹. The value for rate constant with PEG 400 is, therefore 1.40 min⁻¹. Campanelli et al. (1993) found with addition of zinc salt, there was 18% increase of rate constant for a salt concentration 0.1% (w/w PET). The addition of the PEG 400 leads to 3.88 times increase in rate of hydrolysis reaction. Thus, it is concluded that at 240 °C the use of PEG 400 with a very small concentration of 2.11×10^{-5} mol/cm³ results in to extraordinary increment in the rate of PET hydrolysis.

3.3. Influence of reaction conditions on hydrolysis reaction of PET

The reaction mixture consisted of two phases: solid PET and aqueous phase for all experiments and times of reaction. The solid phase was found stuck on the sides and bottom of the reactor for reaction times less than 15 min at 240 °C for an initial reactor loading 22 mol water/mol PET. Similar trend has been observed by Campanelli et al., (1994) with zinc acetate as catalyst. At lower temperatures it was a slurry of PET and aqueous phase and organic mass upon cooling at all temperatures.

3.3.1. Influence of initial molar ratio

The hydrolysis reaction was investigated using PET and water with

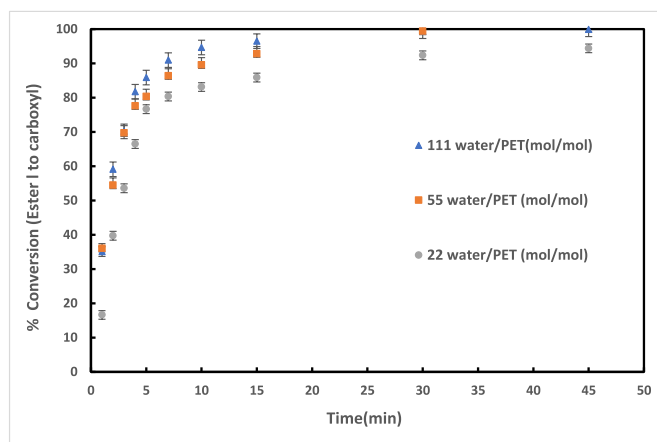


Fig. 5. Carboxyl group concentration with time for three different initial water to PET charge ratio. (Reaction condition- Temperature: 240 °C; Reaction time: 30 min; Catalyst concentration: 2.11×10^{-5} mol/cm³; Speed of agitation: 650 rpm) (Analysis based on end group).

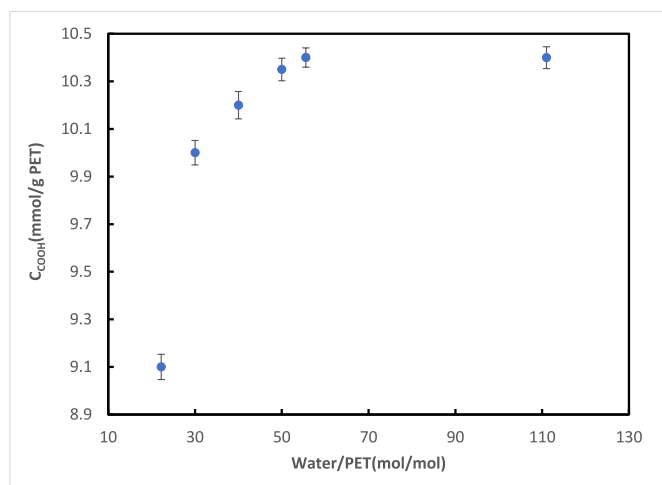


Fig. 6. Equilibrium values of carboxyl group concentration as function of initial water concentration. (Reaction condition- Temperature: 240 °C; Reaction time: 30 min; Catalyst concentration: 2.11×10^{-5} mol/cm³; Speed of agitation: 650 rpm) (Analysis based on end group).

initial charge ratio range of 22–110 mol water/mol PET, Fig. 5 depicts the conversion of carbonyl to carboxylic group (x) as function of time with reaction condition of 240 °C and autogenous pressure (3.2 MPa) generated by water vapour. The initial molar ratio of 22 mol water/mol PET reached 80% conversion within 10 min and thereafter only additional 10% increase in conversion was observed. On the other hand, for initial mol ratio 55–111 mol water/mol PET produced approximately 86–88% conversion in 10 min and reached to 100% conversion within 20 min. The initial mol ratio 55–111 mol water/mol PET gave complete depolymerization, whereas less than 22 mol water/mol PET initial mol ratio reaction failed to give complete depolymerization. Fig. 6 shows the equilibrium concentration of carboxylic group governed by initial molar ratio of water and PET. The equilibrium concentration of carboxyl group does not have impact of PEG 400 rather by initial mol ratio of water: PET and temperature. Campanelli et al. (1993) had reported equilibrium (carboxyl group) formation below 55 mol water/mol PET and it could not reach complete depolymerization. There was an autogenous pressure generation since water was used in large excess than stoichiometrically required. It leads to break down of particles into liquid phase which in turn creates larger surface area helps to form reactive polar intermediate. Also, particle size has major influence on heat and mass transfer, which governs the dissolution of PET in PEG 400. PET depolymerization with reactive extrusion shows the pressure has huge impact on conversion (Yalçinyuva et al., 2022).

Although hydrolysis is investigated using carbonyl conversion to carboxyl group, there might be formation of dimer or trimer of TPA, which are not the desired products, where the dimers and trimers of carboxyl group are considered as TPA. Thus, Figs. S11 and S12 show concentration and percent yield of TPA in reaction mixture, which is the desired product.

3.3.2. Influence of reaction time

The complete depolymerization is majorly governed by reaction time for an all-initial reaction conditions. In Fig. 5, it was observed that within 12 min all ester linkages scission took place. More than 50% reaction time was utilized to achieve the desired product terephthalic acid (TPA), which was also confirmed through percent yield of TPA in Fig. S11. It can also seen from Fig. 5 to reach acid value equivalent to terephthalic acid requires 15–20 min reaction time.

A complete depolymerization is seen in Fig. 5 within 10 min on the basis of ester linkage whereas, Fig. S11 depicts more than 95% yield for TPA on the basis of HPLC analysis in 30 min. This indicates that the

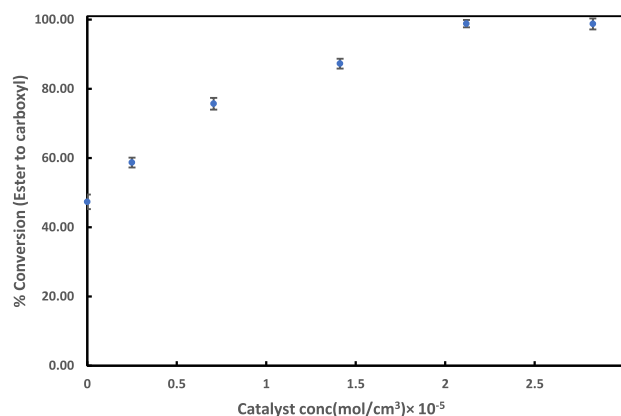


Fig. 7. Effect of PEG 400 various concentration on conversion of ester linkages (%). (Reaction condition- Temperature: 240 °C; PET concentration: 5.27×10^{-4} mol/cm³; Water: 100 cm³; Reaction time: 30 min; Speed of agitation: 650 rpm) (Analysis based on end group).

estimation of conversion based only on ester linkage has limitations and it is required to combine with the analysis of the desired products.

3.3.3. Influence of catalyst concentration on hydrolysis

The influence of PEG 400 concentration on depolymerization was studied in autoclave in the temperature range 230–260 °C, with a reaction time 30 min, for initial reactor loading 111 mol water/mol PET and with $0-3.0 \times 10^{-5}$ mol PEG/cm³. The effect of PEG 400 various concentration on scission of ester linkages to carboxyl group shown in Fig. 7.

Effect of catalyst concentration was studied in a control experiment without any catalyst under similar conditions. The control of experiment (without PEG 400) showed 48 ± 1.65 % conversion of ester linkages to carboxyl group, which was a mixture of oligomers equivalent to the 1 min hydrolysed reaction sample. It is clearly observed that change in catalyst concentration from 1.4×10^{-5} mol PEG 400/cm³ to 2.5×10^{-5} mol PEG 400/cm³ gives sufficient effect about 20% increase in conversion. However, increase of 40% molar concentration leads to more than 80% conversion in scission of ester linkages. Partial increase in conversion does not produce the desired product (TPA); rather it gives a mixture of oligomers, until it reaches complete conversion (carboxyl concentration approx. 10.4 mmol carboxyl group). Thus, molar concentration of PEG 400 of 2.0×10^{-5} mol/cm³ is sufficient to give the maximum conversion in breaking of ester linkages. Increase in the amount of PEG 400 beyond that value does not have any effect on breaking of ester linkages (i.e., carboxyl group content).

Fig. S13 indicates relationship among the yield of TPA at various temperatures using the HPLC basis. The increase in temperature and catalyst concentration increased TPA yield. However, Yu et al. (2012) had reported a decrease in yield above 1.5%, which might be due to the acidic effect of zinc acetate on the stability TPA at high temperature. Among most of the reported catalysts for the PET hydrolysis, glycolysis has shown decrease in yield of TPA with increase in temperature.

Fig. S14 shows that the EG yield increased with increasing concentration of PEG at 220–240 °C. When PEG 400 concentration increased from 0 to 1.5×10^{-5} mol/cm³, the yield of EG increased considerably. After $2-3 \times 10^{-5}$ mol/cm³ the yield remained constant. As against our data, the previous literature reports decrease in yield with increase in temperature as well as catalyst concentration (Sato et al., 2006). However, EG yield had no such effect with PEG 400 in PET hydrolysis in our case.

3.3.4. Effect of catalyst concentration on hydrolysis kinetics

As shown in Fig. 8, the initial rate of reaction was found to increase

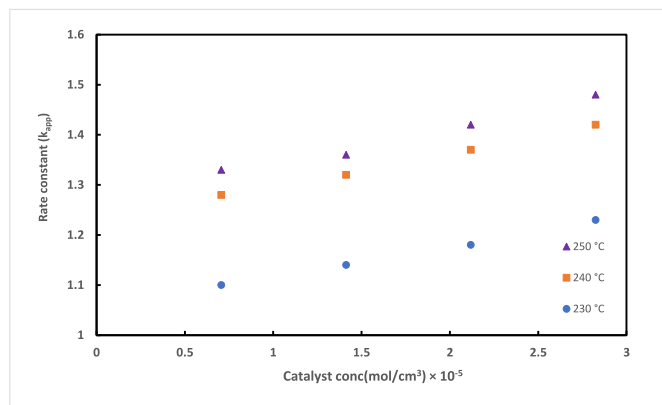


Fig. 8. Effect of PEG 400 concentration on reaction rate constant at different temperatures. (Reaction condition- Temperature: 240 °C; Reaction time: 30 min; PET concentration: 5.27×10^{-4} mol/cm³; Water: 100 cm³; Speed of agitation: 650 rpm) (Analysis based on end group).

Table 1

Effect of Temperature on rate constant (based on end group analysis) for PEG 400 catalysed hydrolysis.

Temperature (°C)	k (min ⁻¹)
230	1.2
240	1.40
250	1.624

Reaction conditions: 55 mol water/mol PET with PEG 400 at M concentration of 2.11×10^{-5} mol/cm³.

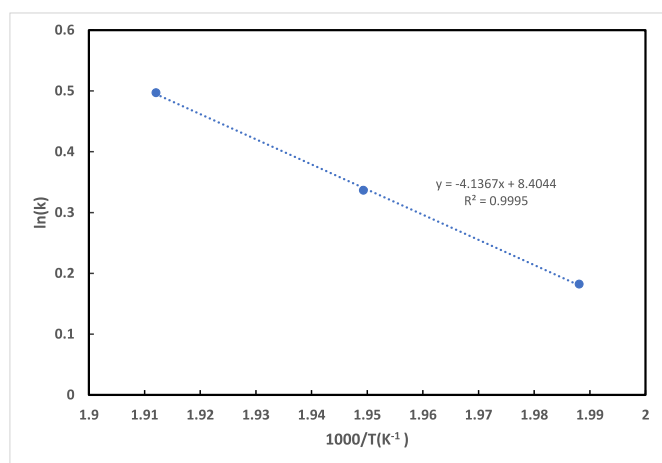


Fig. 9. Arrhenius plot in presence PEG 400 catalytic depolymerization. (Reaction condition- Reaction time: 30 min; Catalyst concentration: 2.11×10^{-5} mol/cm³; PET concentration: 5.27×10^{-4} (mol/cm³) Water: 100 cm³; Speed of agitation: 650 rpm).

with increasing concentration of the catalyst. This is typical of PTC reactions. All further experiments were conducted at a catalyst concentration of 2×10^{-5} mol/cm³.

3.3.5. Influence of temperature on PET hydrolysis

The rate constants for initial charge ratio 55 mol water/mol PET were calculated at different temperatures (Table 1). There could be an onset of mass transfer effects at higher temperature which was explained in the model which is for an instantaneous reaction for which the activation energy values are on lower side. Increase in reaction rate happens in the case of truly kinetically controlled reaction having high activation energy. The Arrhenius plot was made (Fig. 9) to get an apparent

activation energy of 34.4 kJ/mol. The activation energy values can be compared with published literature (Table S2). Uncatalyzed melt phase hydrolysis is reported to have a activation energy as 55.7 kJ/mol (Campanelli et al., 1993) whereas zinc acetate and zinc chloride catalysed hydrolysis with 47.8 (Campanelli et al., 1994) and 64.9 kJ/mol (Wang et al., 2019) respectively. Uncatalyzed high temperature hydrolytic depolymerization of PET on reactive extrusion activation energy is found to be 74.4 kJ/mol (Yalçinyuva et al., 2022). Thus, PEG 400 is the best catalyst in comparison with all previously used catalysts.

The influence of reaction temperature on conversion of carbonyl to carboxyl group was studied, and results are shown in Fig. S15. The result shows that the increase in reaction temperature increases carboxylic group concentration; in other words, it increases initial rate of reaction. Fig. S16 and Fig. S17 depict the effect of temperature for short reaction time. It can be observed an increase in temperature favours the formation of terephthalic acid.

3.4. Catalytic mechanism of PET depolymerization

Examination of PET by TGA and DSC showed that the PET melting temperature is about 240 °C. Then in the melt liquid phase PET reacted with in the aqueous phase containing PEG 400. There is a layer of water around the melt droplets in which the catalyst PEG 400 exists. Scheme 3 depicts the reaction mechanism; water molecule reacts with electron deficient ester linkage which is formed by PEG 400 in repeat unit of PET. PEG 400 hydroxyl group forms a bonding with carbonyl oxygen, which eventually produce electron deficient oxygen ester linkage. Consequently, PET is depolymerised into monomers i.e., TPA and EG. However, the quantitative analysis showed the presence of oligomers or intermediates, which might be dimers or trimers of terephthalic acid. The concentration profile in Fig. 3 indicates that after 3 min, the intermediate oligomers is converted to monomers, i.e., TPA and EG in the presence of PEG 400. Moreover, there might be a possibility of PEG 400 reacting with PET in the melt phase. However, HPLC-ELSD analysis (Fig. 10) depicted the presence of PEG 400 in the aqueous phase, which showed the PEG 400 acts as a phase transfer catalyst which can be recycled. TPA and EG could be recovered easily being separate immiscible phases. The aqueous phase containing PEG was recyclable.

3.5. Catalyst stability and reusability

The aqueous phase containing PEG 400 and EG was made up with water to 100 mL and investigated for reusability for 5 times. Fig. 10 shows the HPLC-ELSD analysis of the reaction liquid phase, which contains PEG 400. It clearly indicates that the ELSD pattern for the standard and reaction liquid mixture show almost similar spectra. However, there is a presence of an additional peak which might be due to the formation of PEG 400-oligomer complex in scattering effect. In addition, there is a little change in the concentration of PEG 400 after the reaction, suggesting that the fidelity of the catalyst structure was maintained during the reaction. Moreover, the catalytic activity of PEG 400 was studied with reuse of reaction liquid phase for subsequent run, which is a solution of EG and PEG 400. As, shown in Fig. S18, recycled liquid phase gave similar activity as that of fresh catalyst for four consecutive recycles.

3.6. Conclusion

Conversion of waste PET plastic into its monomers is highly desirable to achieve circular economy and the net zero goal. A systematic investigation was undertaken to study phase transfer catalysis and a new model for hydrolytic depolymerization of PET was developed. In comparison with the published reports on use of metal acetate and mixed oxide catalysts, PEG 400 exhibited excellent catalytic activity. The limitation of the end group analysis method which was used before is that it does not distinguish among monomer and oligomers. This gap

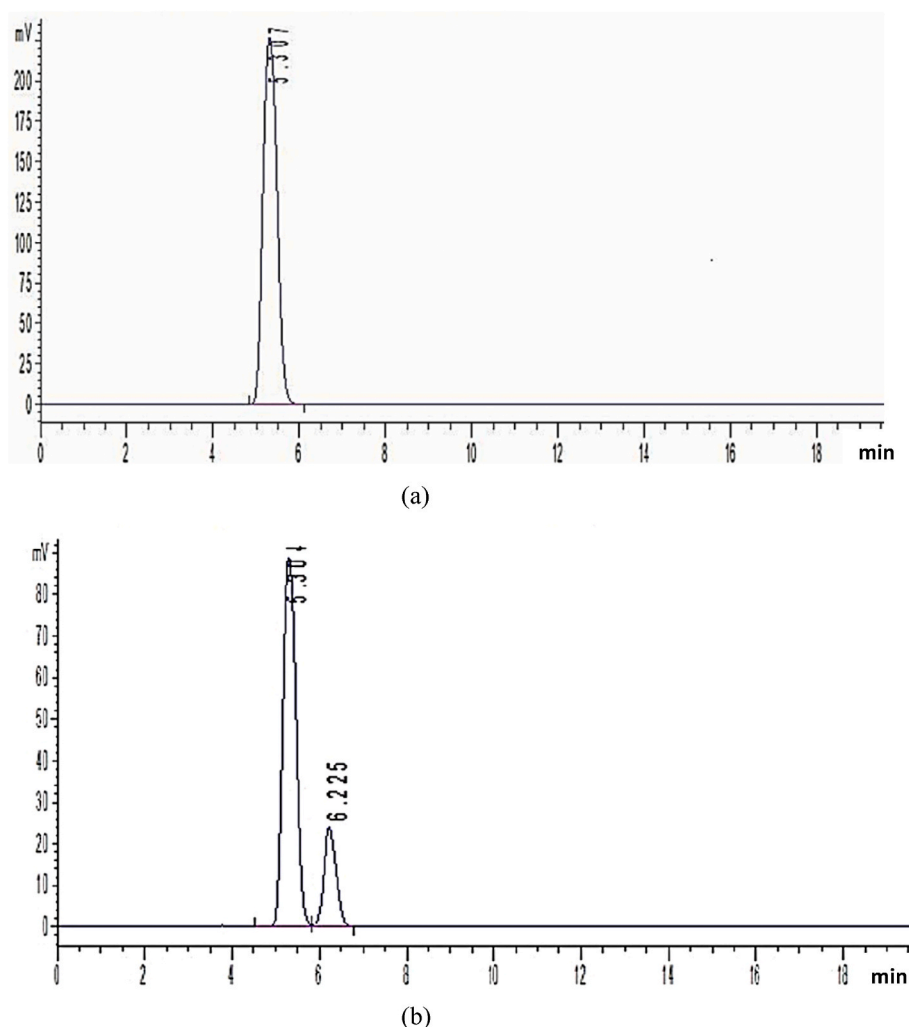


Fig. 10. ELSD plot for (a) standard PEG 400 sample (b) reaction liquid sample after reuse. (Reaction condition- Temperature: 240 °C; Catalyst concentration: 2.11×10^{-5} mol/cm³; PET concentration: 5.27×10^{-4} mol/cm³; Water: 100 cm³; Reaction time: 30 min; Speed of rpm: 650).

was overcome and the concentration profiles of PET, oligomer, and TPA were developed using the HPLC method. The S-L-L-L PTC hydrolysis reaction of PET at melt phase was studied by using PEG 400 as the catalyst. In fact, in melt phase, the so-called S-L PTC turns out to be L-L PTC reaction where the interfacial film is saturated with active intermediate of PEG with PET, which is transferred to aqueous phase for reaction with water than aqueous phase dissolved in solid mentioned in literature. The reaction conditions with initial molar ratio of 55 mol water/mol PET, reaction time 30 min, PEG 400 catalyst concentration 2.0×10^{-5} mol/cm³ at 240 °C, and 3.2 MPa autogenous pressure gave the best yield and conversion of PET. The yield and purity of TPA were found to be 90 and 99.1%, respectively. The synergetic effect of catalyst and pressure generated by water helps increase the conversion and yield of TPA. The addition of PEG 400 as catalyst accelerates depolymerization process by 15 min reaction time in comparison with supercritical and metal acetate depolymerization. In literature most of the PTC reactions were studied either below melt phase or low temperature which required either much higher reaction time or resulted in much lower conversion. The PTC results in the current case are superior to PET depolymerization using zinc salts or that under microwave irradiation with deep eutectic solvent. In the previous studies more than 10% of the reaction product was insoluble in DMSO whereas the entire solid phase was soluble in DMSO in the current work. In other words, all PET is converted to either monomer or lower molecular weight oligomer, which is subsequently converted to monomer and thus conversion of the

oligomer to monomer helps to achieve complete conversion and yield. A pseudo-first-order rate equation was fitted for PET depolymerization with a rate constant of 1.4 min^{-1} at 240 °C, and the apparent activation energy was 34.4 kJ/mol. The rate of hydrolysis is very fast, and complete depolymerization takes place within 30 min. NMR showed excellent monomer purity (terephthalic acid) without product purification or downstream processing. The depolymerization of PET using PEG 400 could be viewed as an eco-friendly process for recycling PET and reuse of monomers.

Author credit statement

Shrirang Sabde conducted all experiments and wrote the first draft and is a Ph.D. student of GDY and RN. The draft was debated, discussed and thoroughly modified by GDY and RN. The mechanism was proposed by GDY. The revision was thoroughly corrected and finalized by GDY. All authors have read and agreed to its publication. Experiments were conducted at ICT Mumbai, India and MSU, East Lansing, MI. SS is a dual Ph.D. degree student of ICT Mumbai and MSU, MI working under the guidance of GDY and RN.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

Appendix A

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2023.138312>.

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